

Grid Dynamics Relevance Brief

How this Physical AI demo maps to GAIN, digital twins, and FDE delivery

One-pager extended brief for screening and stakeholder review

Companion to Full_Project_Report.pdf

FigJam frameworks embedded from project board

Framework source: <https://www.figma.com/board/CmvFbnixCtXsehIEUMbEnZ>

Positioning statement

I built an end-to-end Physical AI pipeline: MuJoCo Aloha simulation, Hugging Face LeRobot ACT for successful bimanual cube transfer, and SmolVLA for language-conditioned actions. I evaluated 50 seeds at 50% CPU success, packaged CI and a release, and documented finetune and SO-100 paths. It follows the same simulate, measure, refine loop Grid Dynamics uses in GAIN Physical AI work.

Delivery loop

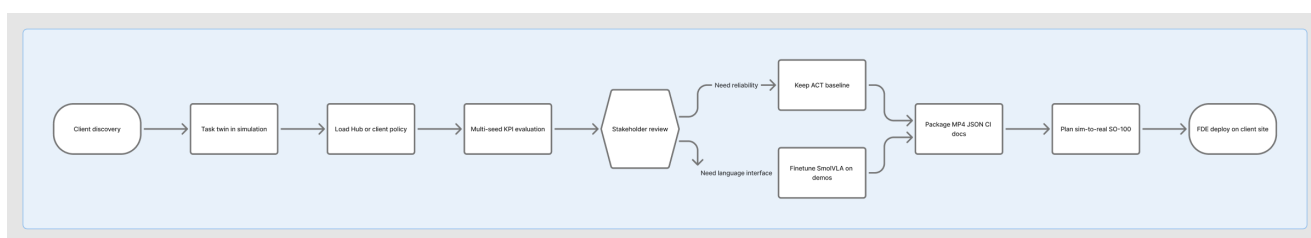


Figure 2. Grid Dynamics GAIN delivery loop (Figma).

Technical stack at a glance

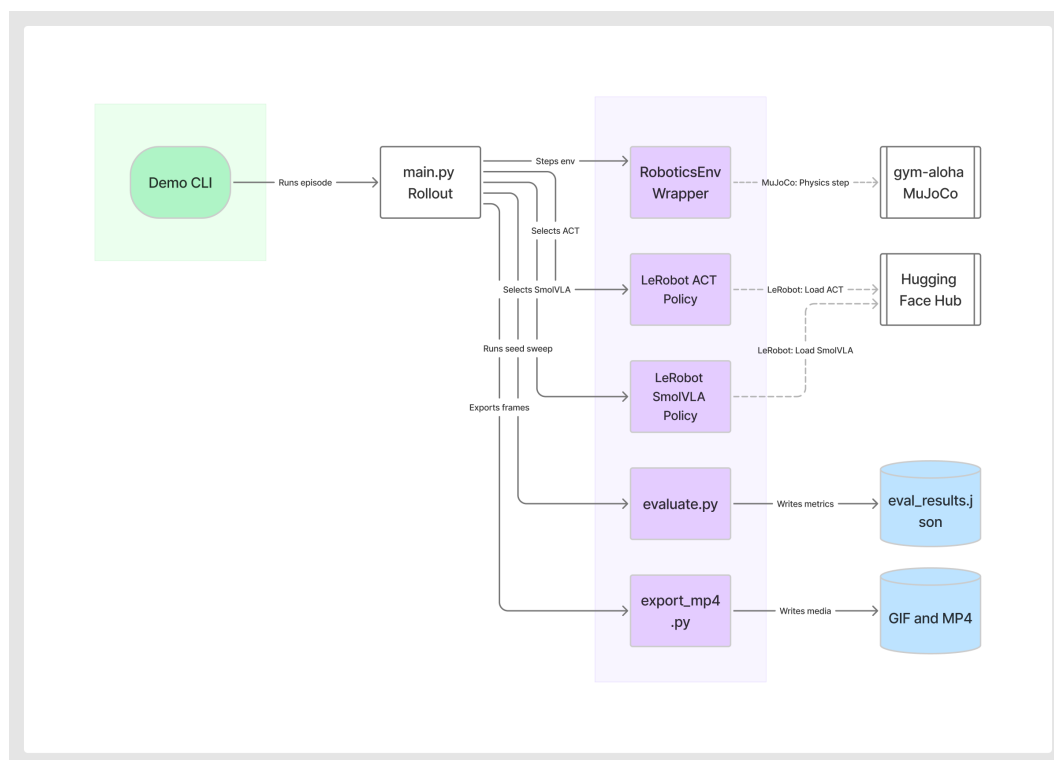


Figure 1. Physical AI system architecture (Figma).

What reviewers should look at

Artifact	Why it matters
demo_output.mp4	ACT solves TransferCube (seed 36)
comparison_act_vs_smolvla.mp4	Reliability vs language interface
eval_results.json	50% multi-seed KPI, not one lucky GIF
prompt_ablation.json	language_sensitive = true
FigJam board	Editable architecture frameworks
docs/FULL_REPORT.md	Full narrative + soft-sensor bridge

Honest boundaries

- Not Omniverse-scale plant twin fidelity.
- Not claiming Aloha ACT zero-shots to SO-100.
- Not claiming laptop CPU matches Hub ~83% GPU eval.
- SmoVLVA success needs finetune; language proof is separate.

Soft-sensor bridge

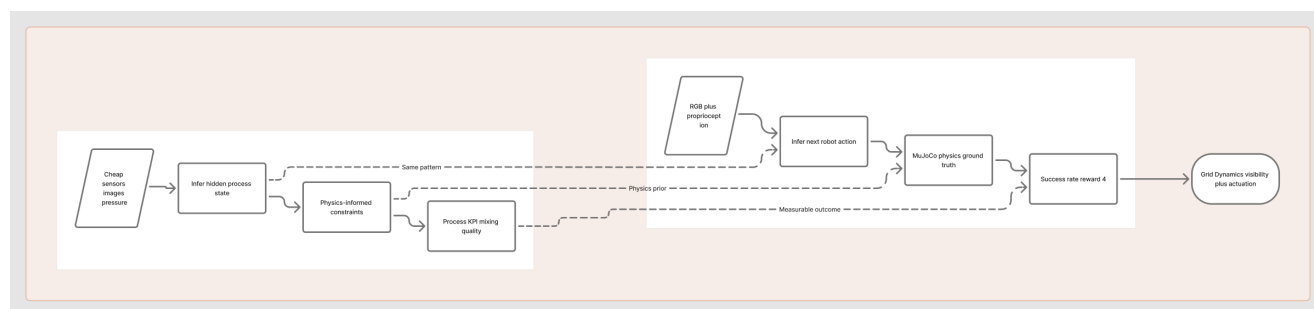


Figure 5. Soft-sensor research mapped to Physical AI (Figma).

Positions the author across perception, estimation, and action: soft sensing for visibility, Physical AI for actuation, digital twins for safe validation.